AI Workflows vs. Agents & MCP Integration Report

Demystifying AI: Workflows, Agents, and the Model Context Protocol

Workflows vs. Agents: Classifying FL-04 In the rapidly evolving AI landscape, the term "agent" is frequently misapplied to any automated process. To build effective systems, we must draw a strict line between a workflow and an agent.

A workflow is a deterministic, pre-defined sequence of operations where the language model and tools are orchestrated through predefined code paths. The path is hardcoded by the developer, who retains complete ownership over the control flow. The AI does not decide what step to take next; it simply executes the prompt it is given at a specific stage and waits for the human or an external script to hand its output to the next stage. Workflows are highly predictable, consistent, and ideal for well-defined tasks where the steps are known upfront. Common workflow patterns include prompt chaining (passing outputs to the next prompt), routing (classifying inputs to direct them to specialists), and evaluator-optimizer loops.

My FL-04 "Source-Grounded Study Notes" pipeline built for the Intelligent Framework For Interpretable Time Series is definitively a workflow. It relies on sequential, human-orchestrated handoffs, mirroring the prompt chaining pattern. I manually upload a PDF to NotebookLM for extraction, manually pass that output to a Claude Project for synthesis, and finally prompt Claude to format a Markdown table. The language model never decides to search for a new paper, nor does it autonomously evaluate if its extraction was successful before moving to the formatting stage. It acts as a highly efficient assembly line, but it is entirely lacking in autonomy.

An agent, conversely, is an AI system where the language model dynamically directs the control flow and its own tool usage. Given an open-ended goal, an agent operates independently in a loop, relying on "ground truth" feedback from its environment to determine which tools to call, in what order, and to evaluate its progress until the objective is met.

Understanding the Model Context Protocol (MCP) An agent is only as useful as the environment it can interact with. This is where the Model Context Protocol (MCP) becomes revolutionary. MCP acts as a universal, open standard—often described as a "USB-C port for AI"—that allows AI applications (MCP hosts) to connect securely to external data sources (MCP servers) without requiring custom integration code for every new service.

MCP relies on a JSON-RPC based data layer featuring three core primitives, each representing a distinct control plane:

Resources: Read-only data sources that provide contextual information, such as file contents, API responses, or database schemas. These are application-

controlled, meaning the host decides when to inject this ambient context into the conversation.

Tools: Executable functions that the language model can autonomously choose to invoke to take action in the real world. Because the model's judgment determines when they are called, tools are the primitive that enables true agentic behavior.

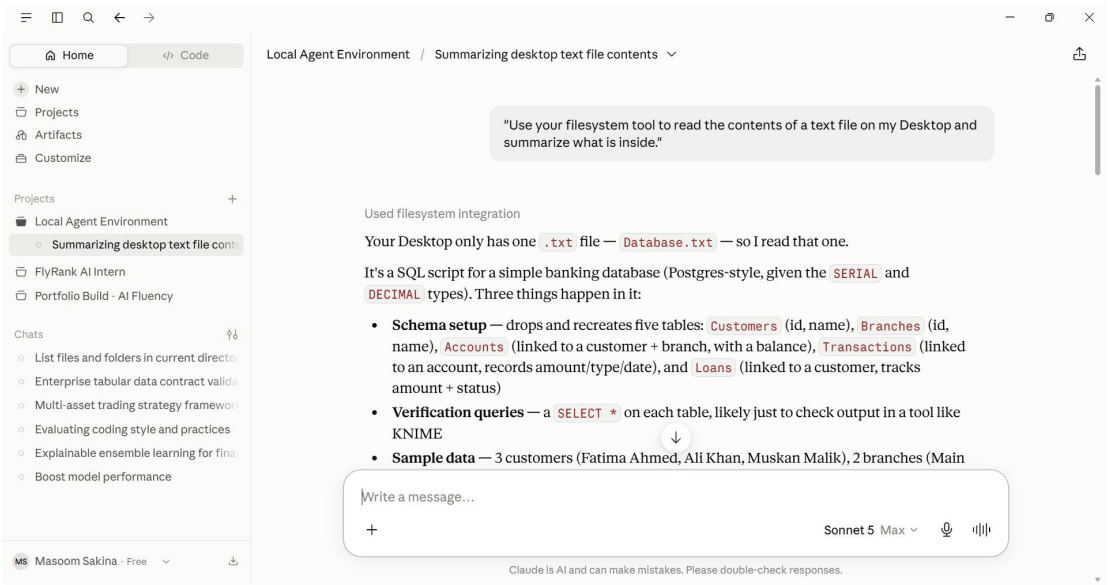
Prompts: Reusable message templates managed by the user. They encode expert workflows and standard instructions, allowing users to trigger complex operations reliably without rewriting instructions from scratch.

Upgrading FL-04 into an Autonomous Agent To transform my FL-04 study notes workflow into a true agent, I would need to integrate it with an MCP server and relinquish control of the routing.

Instead of manually bridging NotebookLM and Claude, I would connect my AI client to a local filesystem MCP and an ArXiv or web search MCP. I would give the agent a single, open-ended overarching command: "Compile a literature review on SHAP explainability in ensemble models. Find five recent papers, extract their methodologies, format the critiques into a Markdown table, and save it to my local 'Capstone_Research' folder."

Operating as an agent, the model would use its internal reasoning to plan the steps. It would autonomously call the search tool to locate relevant PDFs. It would then utilize a parsing tool or resource to read the file contents into its context window. After applying its analytical capabilities to synthesize the critiques, it would finally call a local file-writing tool to generate the final Markdown document. By granting the model the flexibility to dictate its own actions based on environmental feedback, the system transitions from executing rigid, predefined steps to achieving a complex goal autonomously.

Model Context Protocol (MCP) Demonstration.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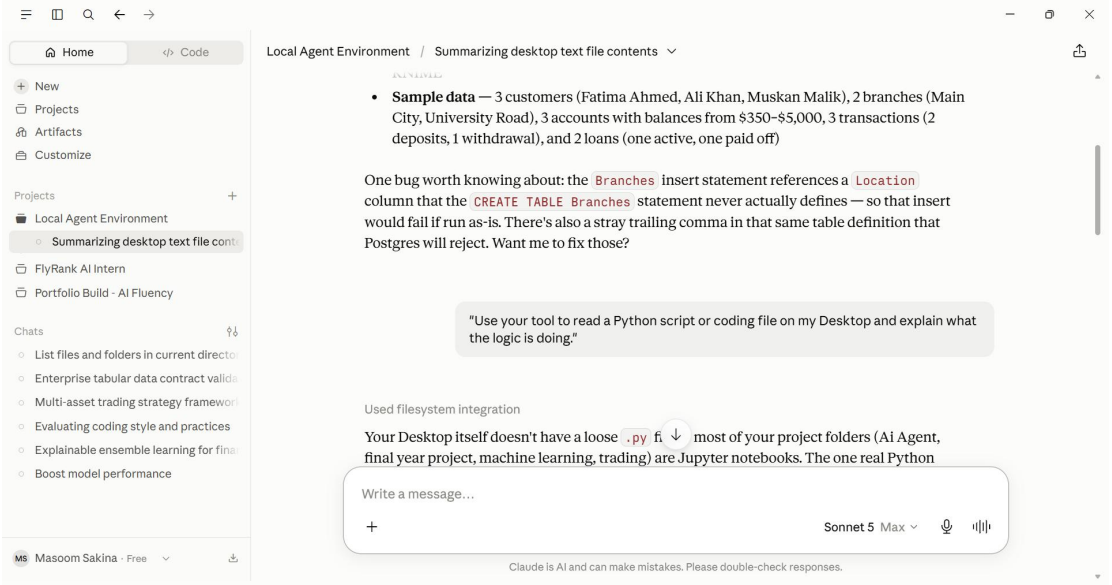
Local Agent Environment / Summarizing desktop text file contents
"Use your filesystem tool to read the contents of a text file on my Desktop and summarize what is inside."
Used filesystem integration
Your Desktop only has one .txt file — Database.txt — so I read that one.
It's a SQL script for a simple banking database (Postgres-style, given the SERIAL and DECIMAL types). Three things happen in it:
Schema setup — drops and recreates five tables: Customers (id, name), Branches (id, name), Accounts (linked to a customer + branch, with a balance), Transactions (linked to an account, records amount/type/date), and Loans (linked to a customer, tracks amount + status)
Verification queries — a SELECT * on each table, likely just to check output in a tool like KNIME
Sample data — 3 customers (Fatima Ahmed, Ali Khan, Muskan Malik), 2 branches (Main
Write a message...
Sonnet 5 Max
Claude is AI and can make mistakes. Please double-check responses.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Sample data — 3 customers (Fatima Ahmed, Ali Khan, Muskan Malik), 2 branches (Main City, University Road), 3 accounts with balances from $350–$5,000, 3 transactions (2 deposits, 1 withdrawal), and 2 loans (one active, one paid off)
One bug worth knowing about: the Branches insert statement references a Location column that the CREATE TABLE Branches statement never actually defines — so that insert would fail if run as-is. There's also a stray trailing comma in that same table definition that Postgres will reject. Want me to fix those?
"Use your tool to read a Python script or coding file on my Desktop and explain what the logic is doing."
Used filesystem integration
Your Desktop itself doesn't have a loose .py fi most of your project folders (Ai Agent, final year project, machine learning, trading) are Jupyter notebooks. The one real Python
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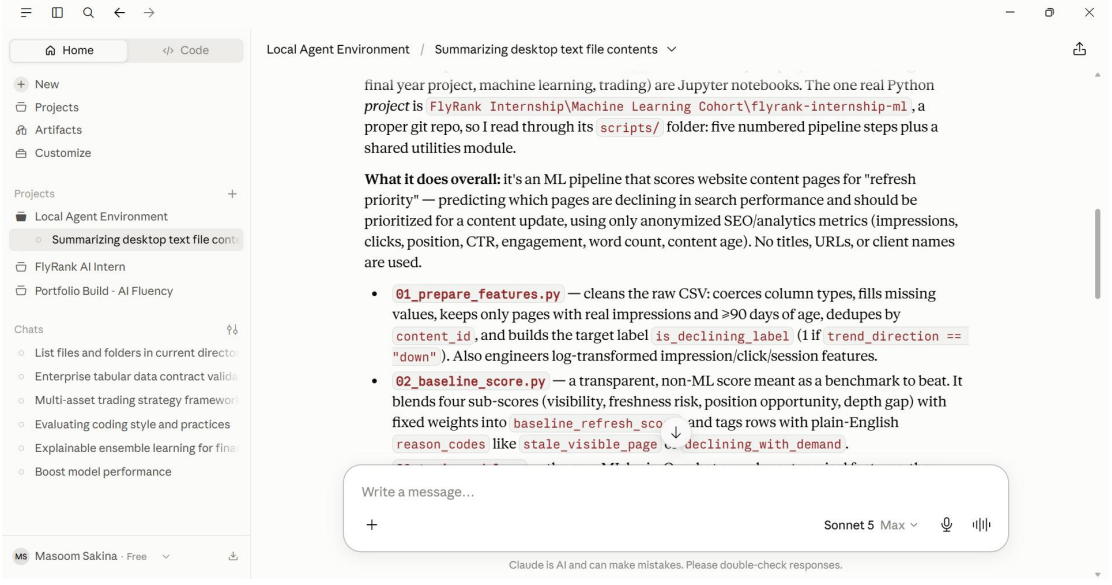
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final year project, machine learning, trading) are Jupyter notebooks. The one real Python project is FlyRank Internship\Machine Learning Cohort\flyrank-internship-ml, a proper git repo, so I read through its scripts/ folder: five numbered pipeline steps plus a shared utilities module.
What it does overall: it's an ML pipeline that scores website content pages for "refresh priority" — predicting which pages are declining in search performance and should be prioritized for a content update, using only anonymized SEO/analytics metrics (impressions, clicks, position, CTR, engagement, word count, content age). No titles, URLs, or client names are used.
01_prepare_features.py — cleans the raw CSV: coerces column types, fills missing values, keeps only pages with real impressions and ≥90 days of age, dedupes by content_id, and builds the target label is_declining_label (1 if trend_direction == "down"). Also engineers log-transformed impression/click/session features.
02_baseline_score.py — a transparent, non-ML score meant as a benchmark to beat. It blends four sub-scores (visibility, freshness risk, position opportunity, depth gap) with fixed weights into baseline_refresh_sco and tags rows with plain-English reason_codes like stale_visible_page declining_with_demand.
Write a message...
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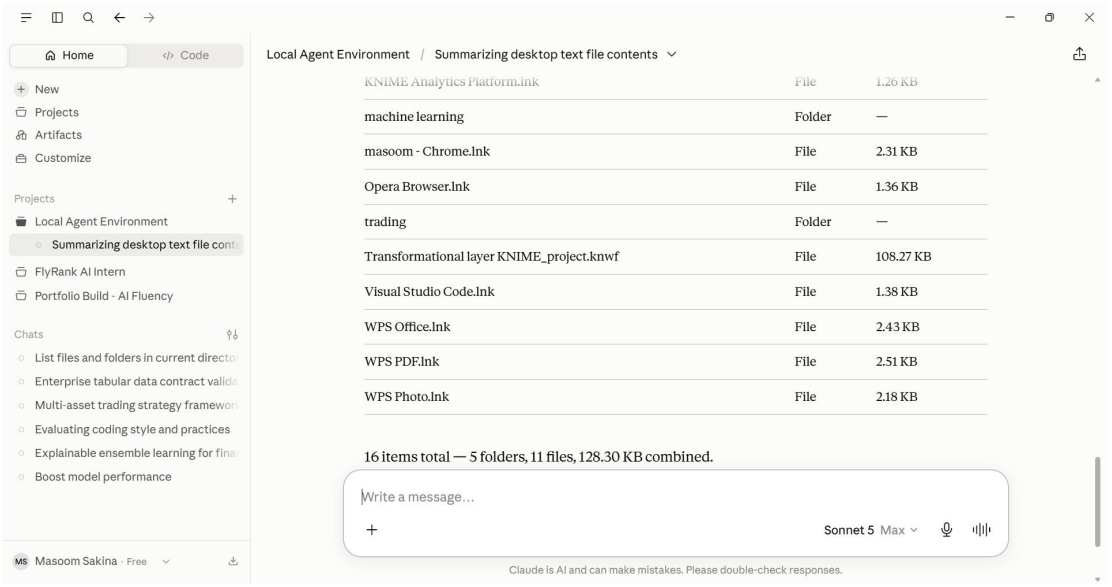
- `03_train_model.py` — the core ML logic. One-hot encodes categorical features, then splits by *client* rather than by row when possible (`make_client_aware_split`) so the model is tested on clients it's never seen — avoiding leakage — before falling back to a stratified row split. Trains and compares Logistic Regression, a Decision Tree, and a Random Forest, picks the best by `precision_at_50` (how many of the top 50 ranked pages are true declines), then retrains that model on all the data and saves its feature importances.
- `04_evaluate_and_export.py` — blends the model's probability with the baseline score (70/30) into a single 0–100 `final_refresh_score`, assigns a `confidence` tier and a `suggested_action` (`expand_and_refresh`, `refresh`, `monitor`, etc.), ranks everything into a final queue CSV, and generates SVG bar charts plus a Markdown report.
- `05_build_pdf_report.py` — pure presentation. Takes the JSON/CSV from the previous step and lays out a branded, multi-page PDF (KPI cards, metric tables, hand-drawn bar charts via a custom `ReportLab` `Flowable` subclass) — no new analysis happens here.
- `ml_utils.py` — the shared backbone imported by everything else: path constants, the canonical feature-column lists, and helpers like `normalize`, `percentile_rank`, `precision_at_k`, and a dependency-free SVG chart writer.
- `run_all.py` — the orchestrator. Just runs the five numbered scripts in order as subprocesses and prints a final summary once `summary.json` exists.

The clever bits are in steps 3 and 4: a client-aware train/test split to keep evaluation honest, three models judged by precision at the top of the ranked list (the metric that actually matters for a prioritized queue) rather than plain accuracy, and a final score that deliberately blends ML output with an interpretable rule-based baseline instead of trusting the model alone. The code is explicit throughout that this is a reviewer aid, not an auto-publish decision — the report and PDF both call out that high-confidence rows still need manual review.

"Use your tool to list all the files currently sitting on my Desktop and format them into a Markdown table."

List Directory with Sizes

Name	Type	Size
Ai Agent	Folder	—
ALL.sql	File	3.16 KB
Database.txt	File	3.16 KB
desktop.ini	File	282 B
final year project	Folder	—
FlyRank Internship	Folder	—
KNIME Analytics Platform.lnk	File	1.26 KB
machine learning	Folder	—
masoom - Chrome.lnk	File	2.31 KB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KNIME Analytics Platform.lnk File 1.26 KB
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masoom - Chrome.lnk File 2.31 KB
Opera Browser.lnk File 1.36 KB
trading Folder —
Transformational layer KNIME_project.knwf File 108.27 KB
Visual Studio Code.lnk File 1.38 KB
WPS Office.lnk File 2.43 KB
WPS PDF.lnk File 2.51 KB
WPS Photo.lnk File 2.18 KB
16 items total — 5 folders, 11 files, 128.30 KB combined.
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